Exercise Session #5

Potentiometer, Light Sensor and Macros

Potentiometer · Disco in Darkness · Macros vs. constexpr

Overview

- → Setup
- → Exercise 1: Reading the potentiometer
- → Exercise 2: Disco in Darkness
- → Exercise 3: Macros vs. constexpr
- → Homework



Setup

→ Wire up the Pico W and the Debug Probe

→ Remember how to get into BOOTSEL

→ Pull the latest from the PHCE 2026 repository

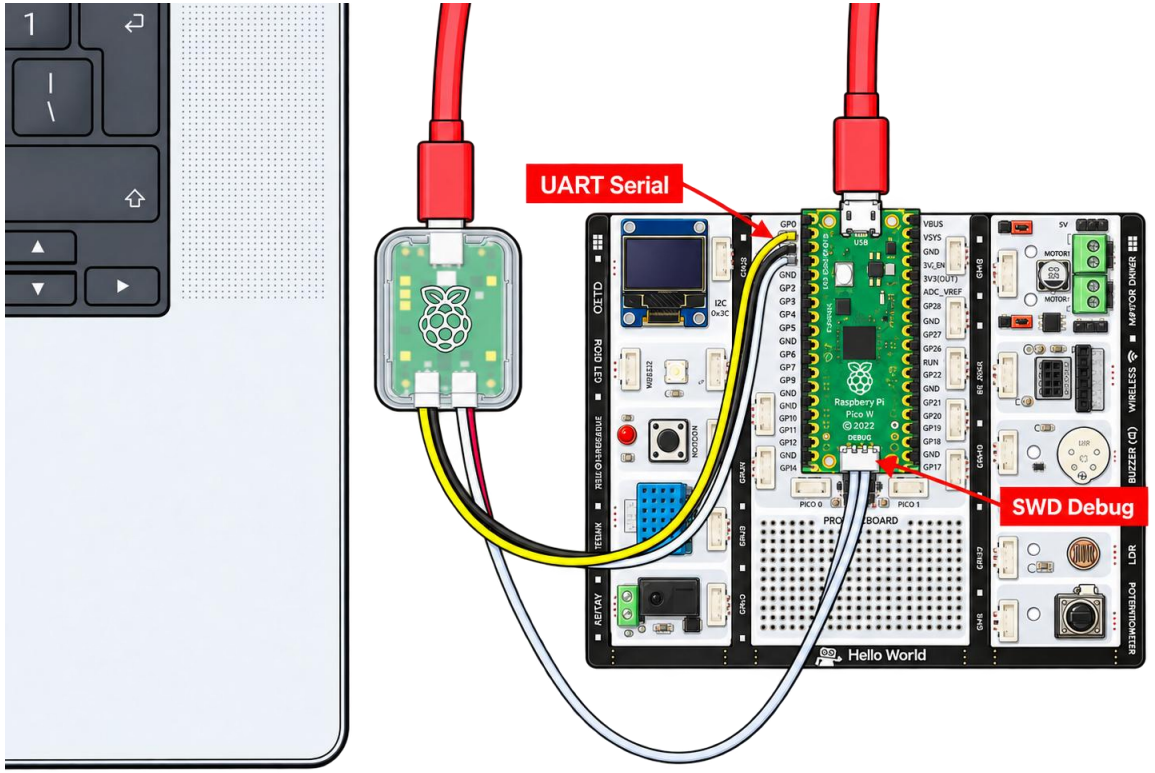
Hardware Connection

→ Connect Pico W

→ Connect Debug Probe

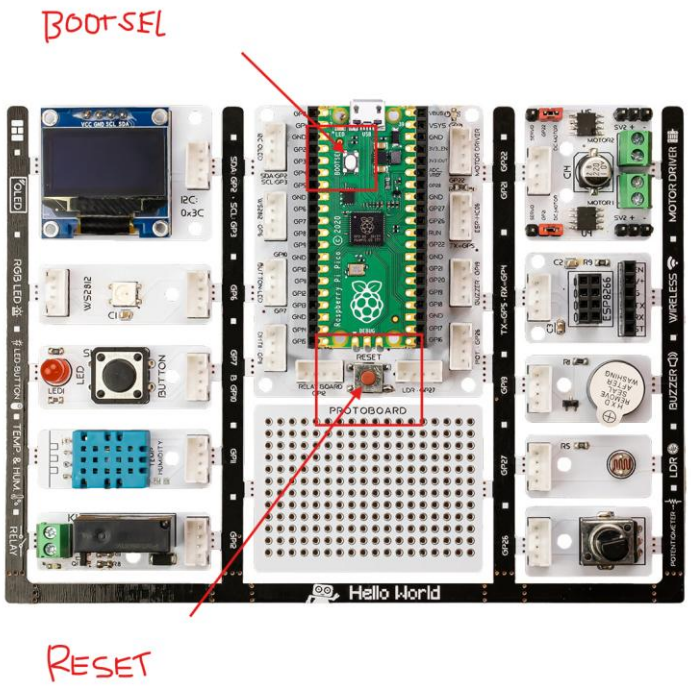
→ Check UART wiring:

- Yellow: TX
- Red / Orange: RX
- Black: GND



Hardware

You don’t have to unplug the Pico



To enter BOOTLOADER:

1. Press and Hold “RESET”
2. Press and Hold “BOOTSEL”
3. Let go of “RESET”
4. Let go of “BOOTSEL”

Software

We still use the PHCE 2026 repository

```
git fetch origin
git checkout <your-branch> // if not already in your branch
git merge origin/master
```

Run this before every exercise. Skip the checkout if you are already on your branch.

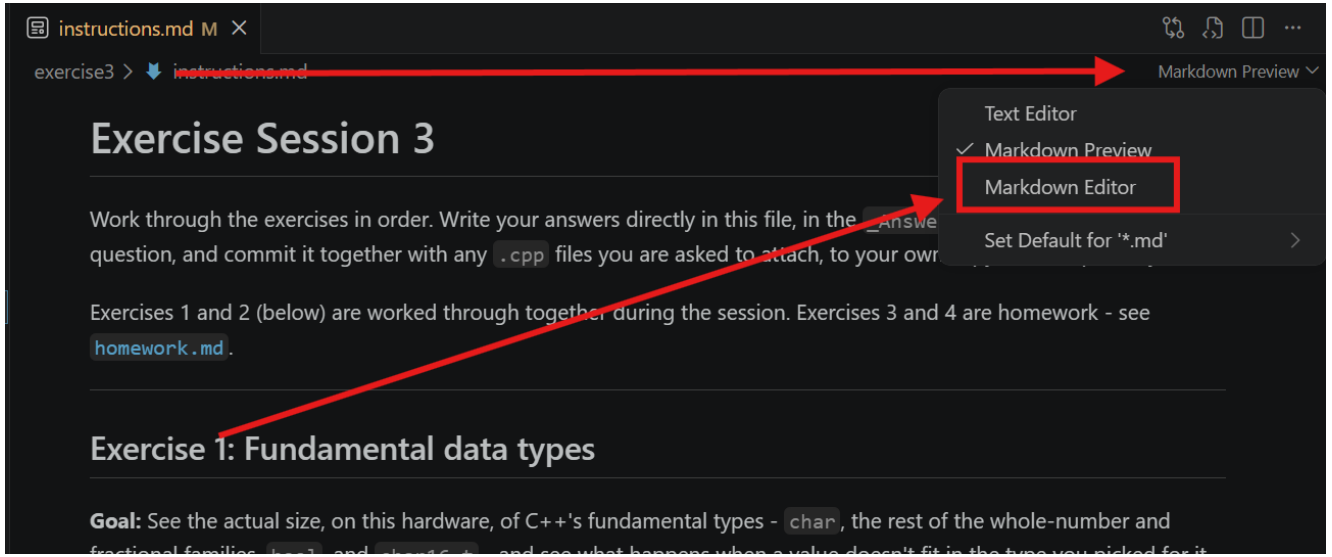
source.coderefinery.org/sdm-edu/phce-2026

Writing Answers in VS Code

→ Open `exercise5/instructions.md`

→ Top right: **Markdown Preview → Markdown Editor**

→ Type your answer straight into the preview, under **Answer:**



Markdown Editor: edit the rendered file directly, no raw markdown

On to the exercises

Exercise 1: Analog Input

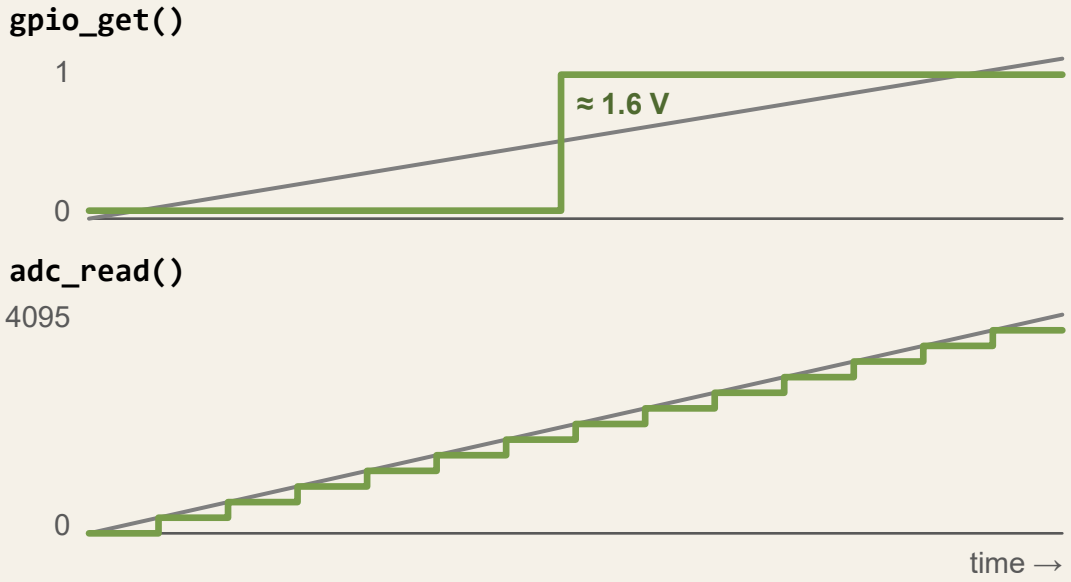
→ **Digital** pins only know **0 or 1**

→ The **ADC** measures the voltages in between

→ **12 bits**: 0–3.3 V becomes **0–4095**

→ One step is **0.8 mV**, so noise **moves the last bits**

Same pin, two ways to read it



The digital pin sees one jump. The ADC sees every step.

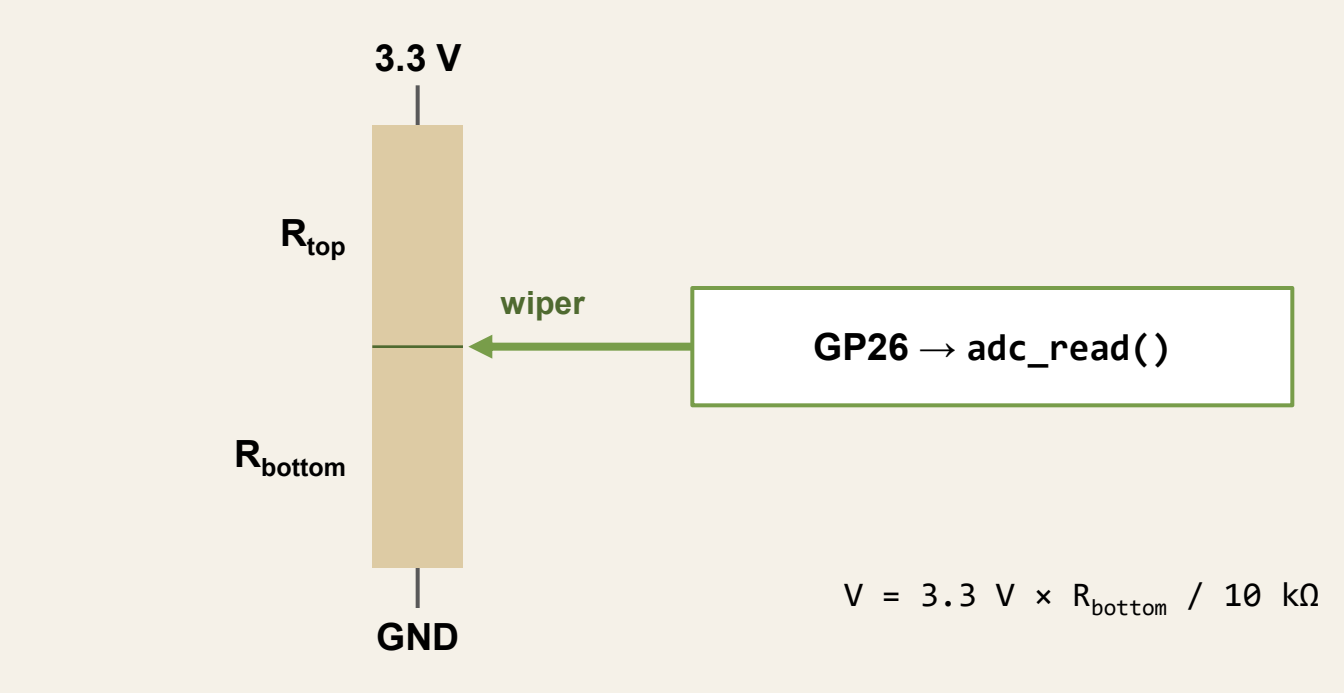
The Potentiometer Is a Voltage Divider

→ Three pins: **3.3 V**, **GND**, and the **wiper** on GP26

→ The knob **slides the wiper** along a 10 kΩ track

→ The wiper picks off a **share of 3.3 V**

→ Knob halfway → **1.65 V** → about **2048**



Exercise 1: Your Turn

15 min

Ask if you need help

- → Fill in `raw_to_volts()` and `raw_to_ohms()`
- → Turn the knob fully both ways and fill in the table
- → Hold the knob still: how much does `raw` jump? And the 8-bit value?
- → Answer the questions in `instructions.md`

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Break (10mins)



Exercise 2: Disco in Darkness

The light sensor is a voltage divider like the pot, with a light-dependent resistor as one half.

1 · Light sensor

less light → higher voltage on GP27

2 · ADC

`adc_select_input(1)` → `raw`

3 · `main()`

compares `raw` with your threshold → LED on or off

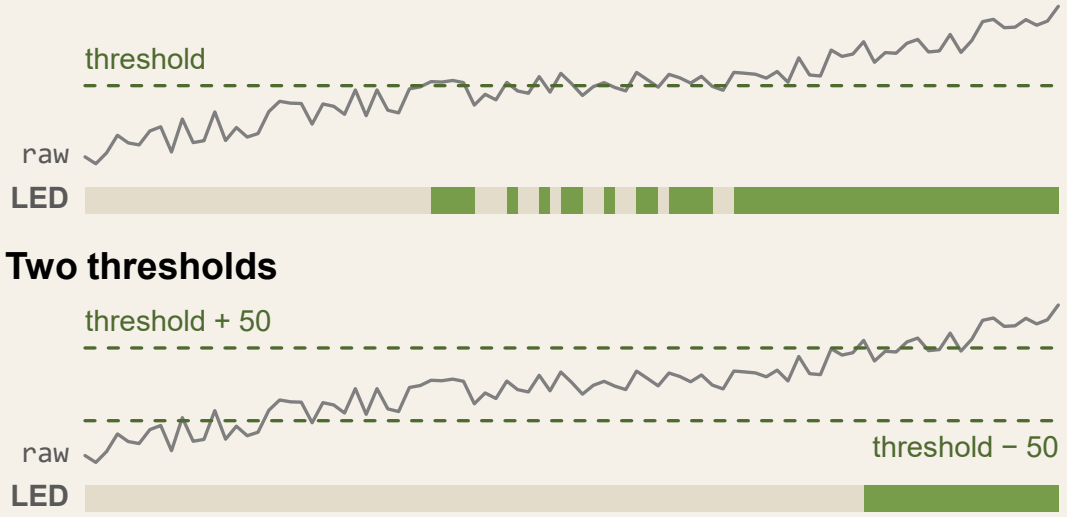
Measure your own numbers first. Every sensor and every room is different.

Why Two Thresholds

→ Near the threshold, `raw` **wobbles** a few steps

→ Every wobble across the line **flips the LED**

→ Fix: **two thresholds**: on 50 above yours, off 50 below



The LED only changes when the reading really moved.

Exercise 2: Your Turn

15 min

Ask if you need help

- → Turn the LED on in the dark, with two thresholds around your own number
- → Half cover the sensor: the LED stays steady
- → Disco: RGB LED on GP6, then cycle the colors
- → Answer the questions in `instructions.md`

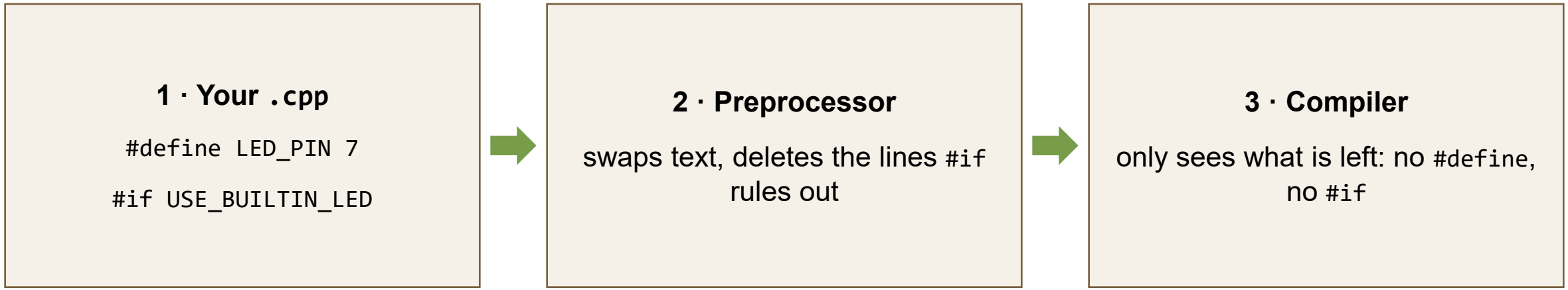
source.coderefinery.org/sdm-edu/phce-2026

Break (10mins)



Exercise 3: The Preprocessor Runs First

Before the compiler sees your code, the preprocessor edits it as text.



The preprocessor doesn't know C++. It doesn't know types, scope or * before +.

Macros vs. constexpr

```
#define LED_PIN 7             // text, swapped everywhere
constexpr uint LED_PIN {7};  // a real constant: type, scope

#define DEBUG_MODE
#ifdef DEBUG_MODE
  #define DEBUG_LOG(t, v) std::cout << t << v << std::endl
#else
  #define DEBUG_LOG(t, v)    // becomes nothing
#endif
```

`#define` constants: no type, no scope. Use **`constexpr`** instead.

`#if` / **`#ifdef`**: the preprocessor's good job. It picks which code exists.

A macro gets **text**, a function gets a **value**. Try `DOUBLE(3 + 4)`.

Exercise 3: Your Turn

20 min

Ask if you need help

- → Turn the `#define` constants into `constexpr`
- → Find out why `DOUBLE(3 + 4)` isn't 14
- → Swap `#if` for a plain `if` and read the error
- → Break the build on purpose with `EXERCISE`, then turn off `DEBUG_MODE`
- → Answer the questions in `instructions.md`

source.coderefinery.org/sdm-edu/phce-2026

Homework

→ Homework in `exercise5/homework.md`: **swap two numbers** (value, pointer, reference) and a **function that calls itself**

→ Step 6 crashes the board on purpose. That's the point.

Thank you 